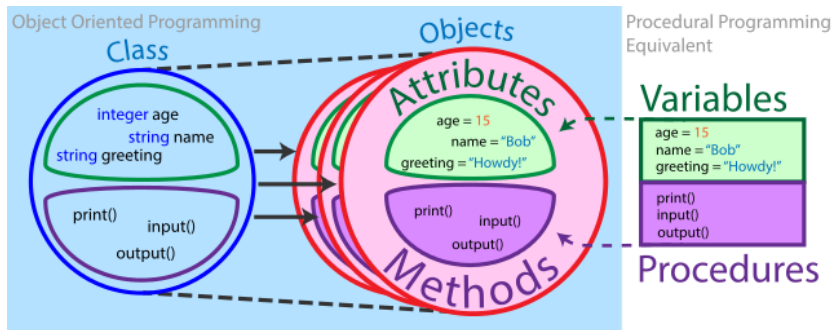


Object Oriented Software Principles and Design



Computer Science CS 611
Boston University

Christine Papadakis-Kanaris

What is Computer Science?

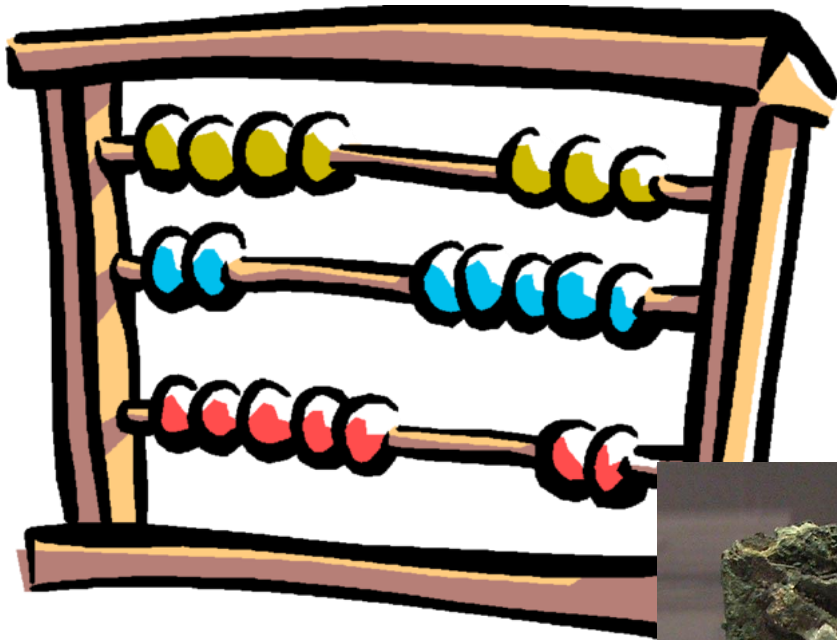
Computer Science is an embodiment of human intelligence and ingenuity. It is the byproduct of our quest to understand the universe, our need to find solutions to known problems, and a desire to free us from our own physical limitations.

Milestones in Computer Science

If I were to ask you to name one of the first important developments on the road to the modern-day computer, what would you say?

Inventions along the way...

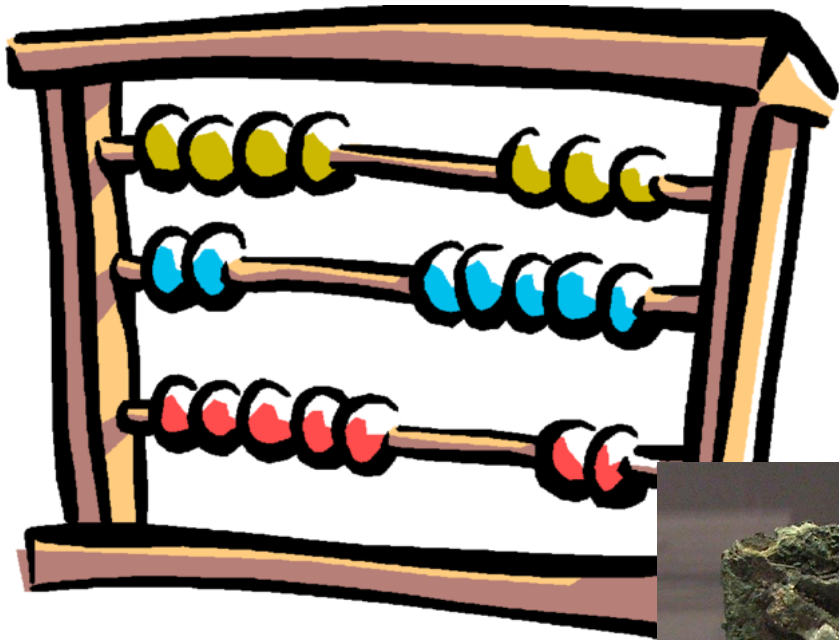
Antikythera Machine (2nd-1st century B.C.)



The Antikythera
Wreck

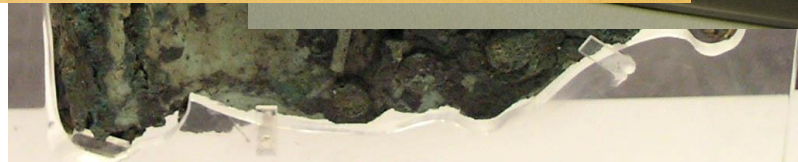
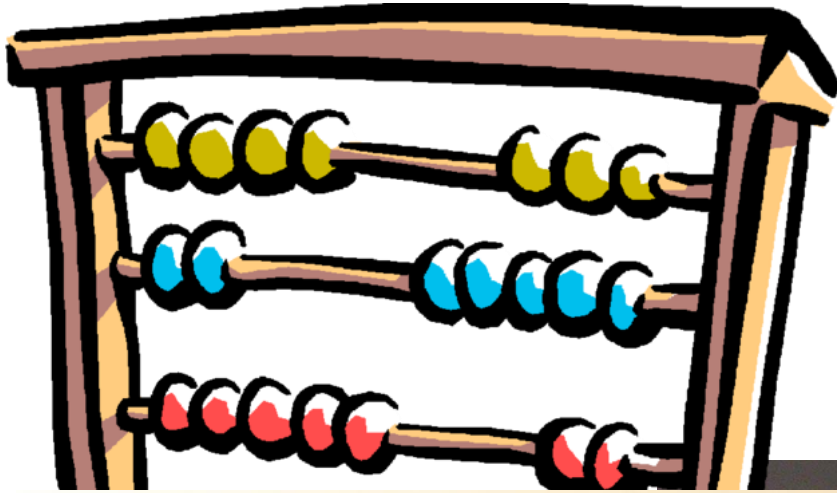
Inventions along the way...

Pascal's Calculator (mid 17th century)



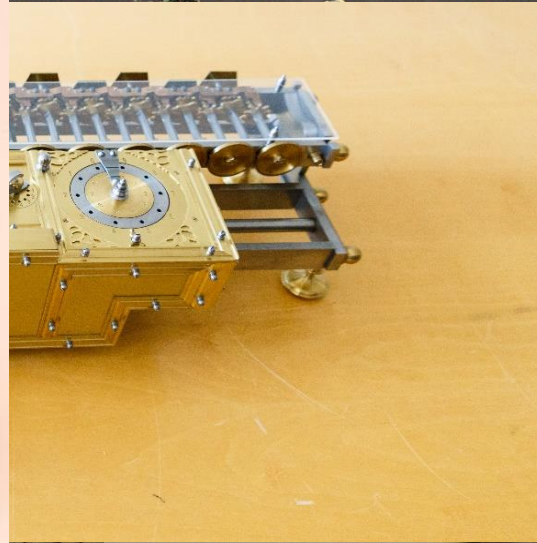
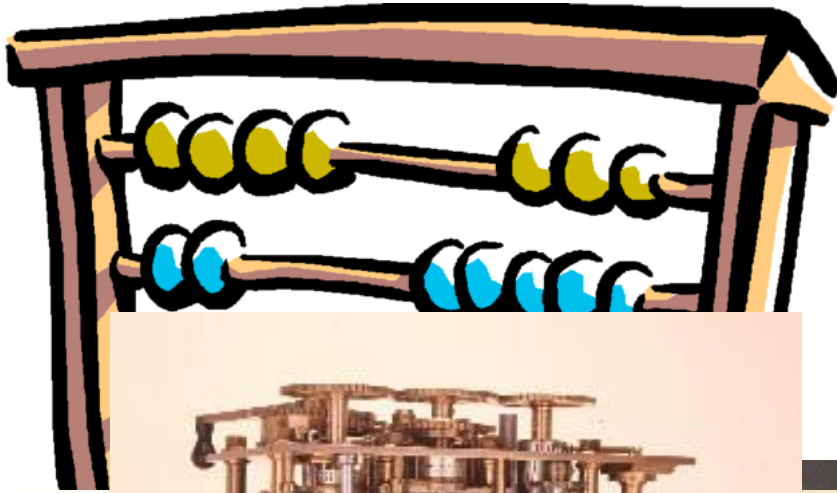
Inventions along the way...

Leibniz wheel (mid 17th century)



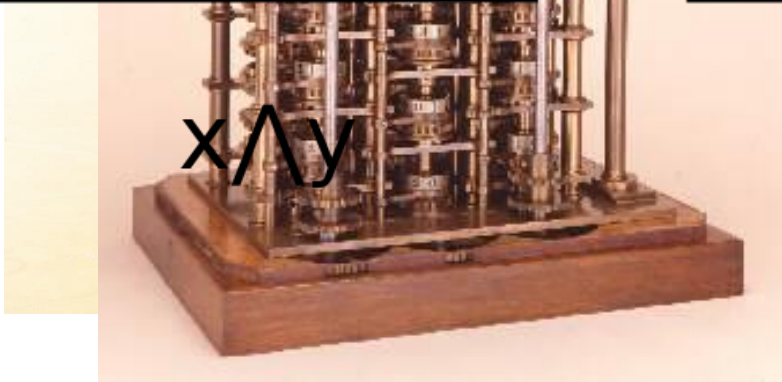
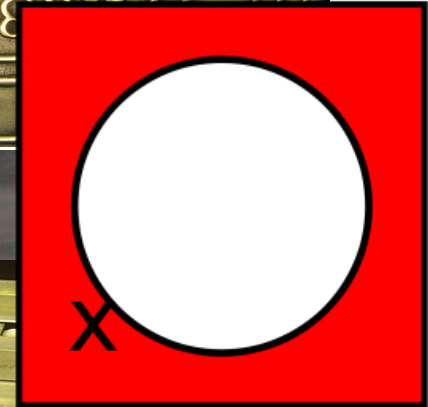
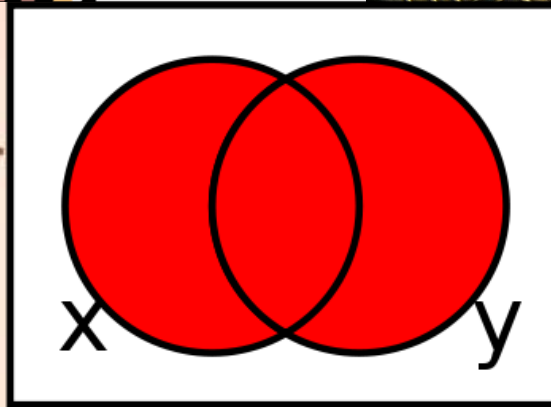
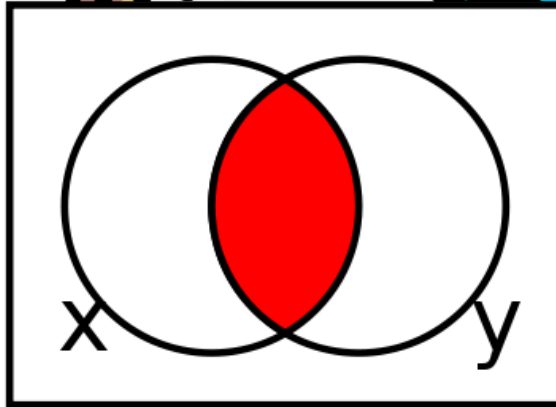
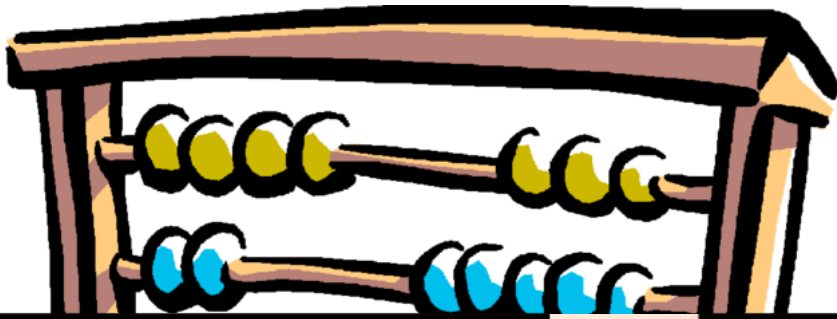
Inventions along the way...

Difference Engine (early 19th century)

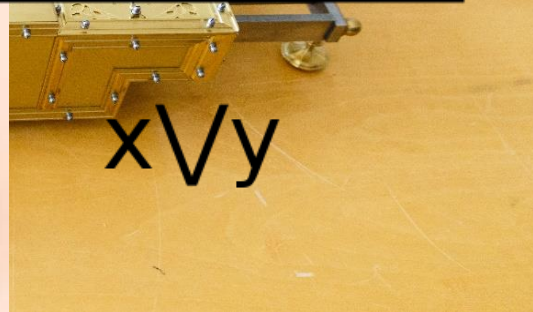


Inventions along the way...

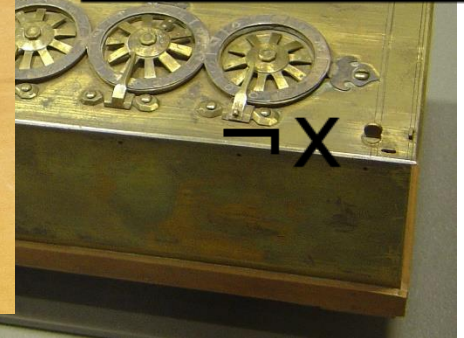
Boolean Logic



$$x \wedge y$$



$$x \vee y$$



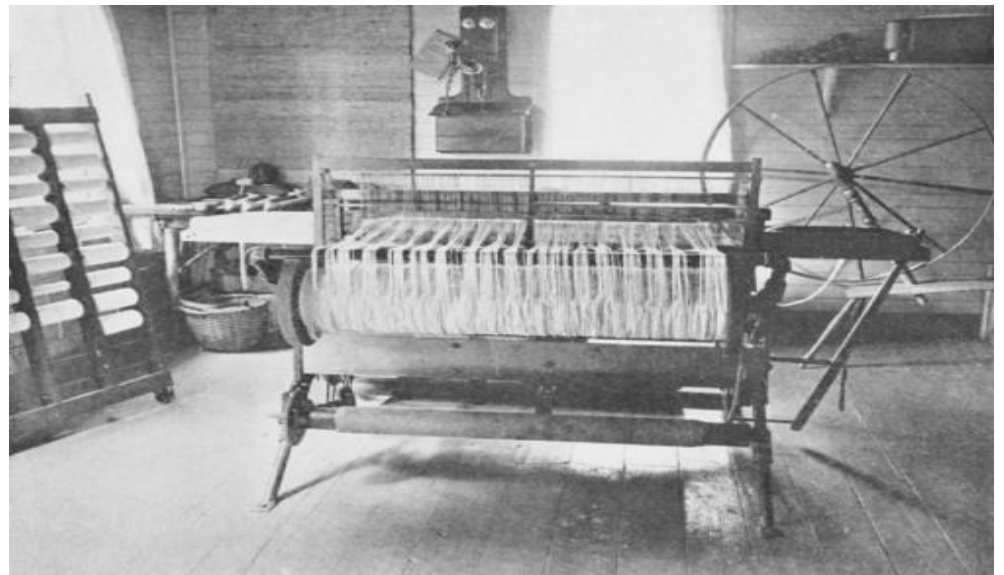
$$\neg x$$

Milestones in Computer Science

If I were to ask you to name one of the first important developments on the road to the modern-day computer, what would you say?

How many of you would believe that it came out of the textile industry?

Well, it did!!!



Milestones in Computer Science

1801: In France, Joseph Marie Jacquard invents a loom that uses **punched wooden cards** to automatically weave fabric designs.

[Jacquard Loom
in action](#)

Why is this significant?

Milestones in Computer Science

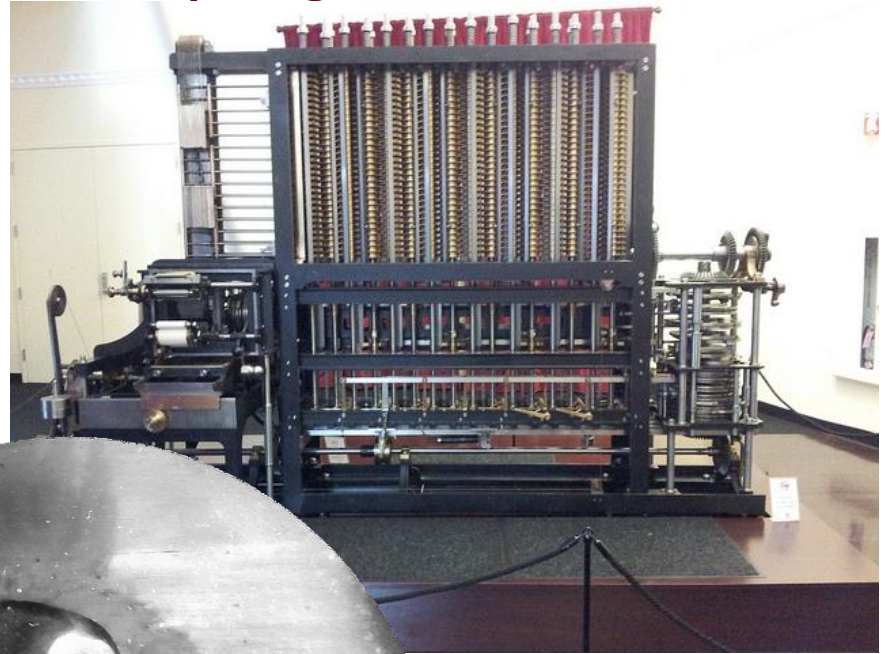
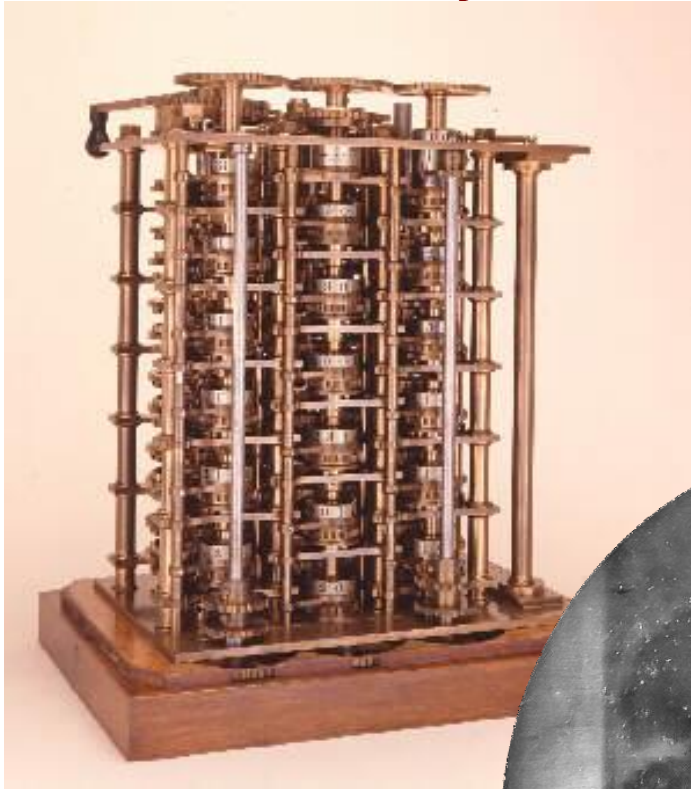
1801: In France, Joseph Marie Jacquard invents a loom that uses **punched wooden cards** to automatically weave fabric designs.

Why is this significant?

In doing so, Joseph Marie Jacquard invents the first “**programmable**” device.

Augusta Ada Lovelace

Lady Lovelace, *first programmer*



[Lady Lovelace's notes on the Analytical Engine](#)

Milestones in Computer Science

1801: In France, Joseph Marie Jacquard invents a loom that uses **punched wooden cards** to automatically weave fabric designs.

Why is this significant?

In doing so, Joseph Marie Jacquard invents the first “programmable” device.

Who would believe that another major milestone came from a statistician working for the U.S. census bureau?

Well, it did!

1890: Herman Hollerith designed and built programmable card processing machines used to calculate the 1890 census, accomplishing the task in just two years and saving the government millions!



[Hollerith
Tabulator](#)

And the future (as we know it)
is established!

You have to believe that anyone who
completes a task years ahead of
estimates **saving \$\$\$\$** is going to get
noticed. And he was!

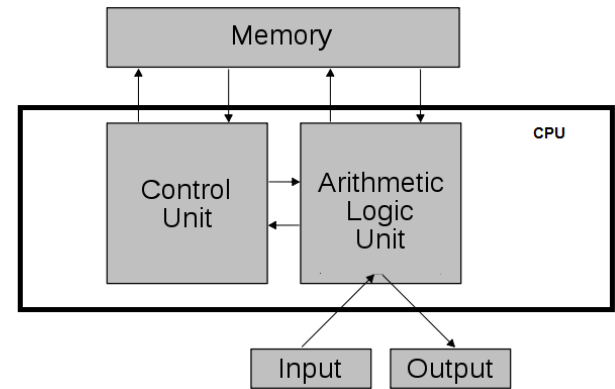
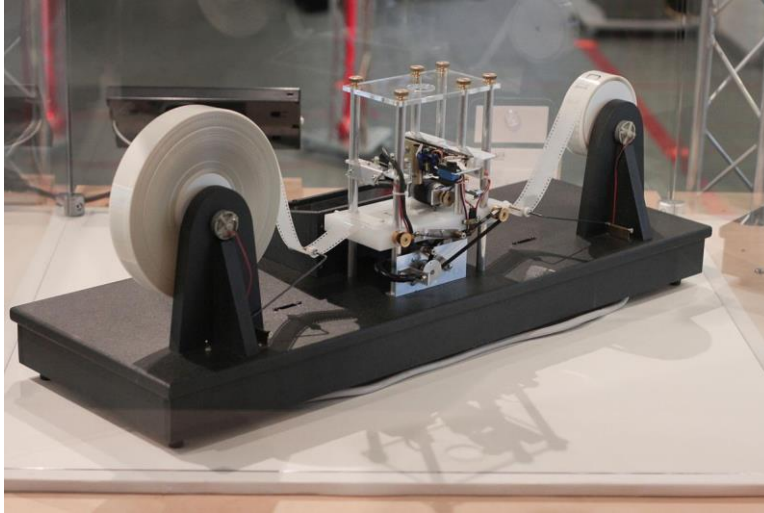


Together with several
investors, Herman Hollerith
establishes a little company
which ultimately became
known as....

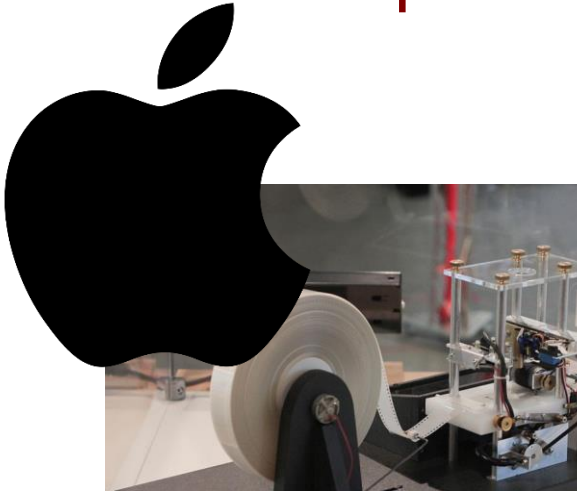
IBM!!!

The exponential growth of the 20th century:

Von Newman Architecture



The exponential growth of the 20th century



THE
C
PROGRAMMING
LANGUAGE



M. Ritchie

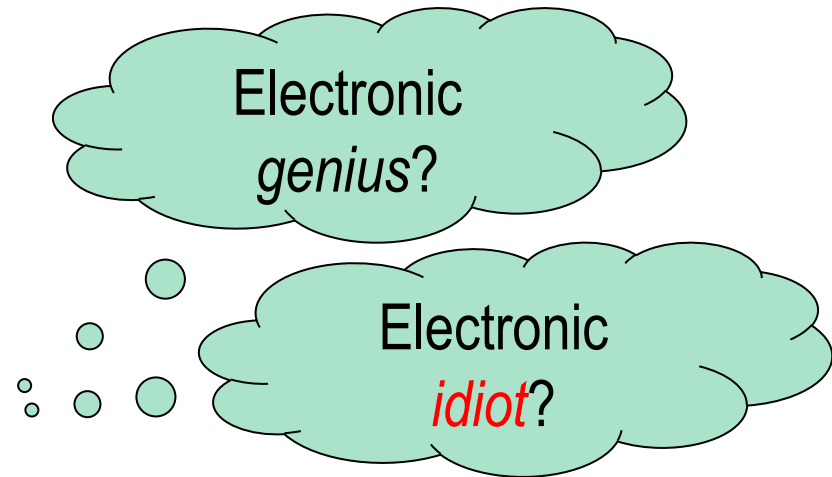
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in mon

History on
Computing

Fundamental Question in Computer Science

- Are computers intelligent?



Fundamental Question in Computer Science

- Are computers intelligent?
- A computer is just a **black box**

A (binary) device that
responds to
two types of signals
on and off!

- *A symbol processing machine which does exactly what we program it to do – nothing more and nothing less.*



Fundamental Question in Computer Science

- Are computers intelligent?
- A computer is just a **black box**

A (binary) device that
responds to
two types of signals
on and off!

*It is the layers of **software**
executing on a computer
that make computers
interesting and give the
illusion of intelligence!*

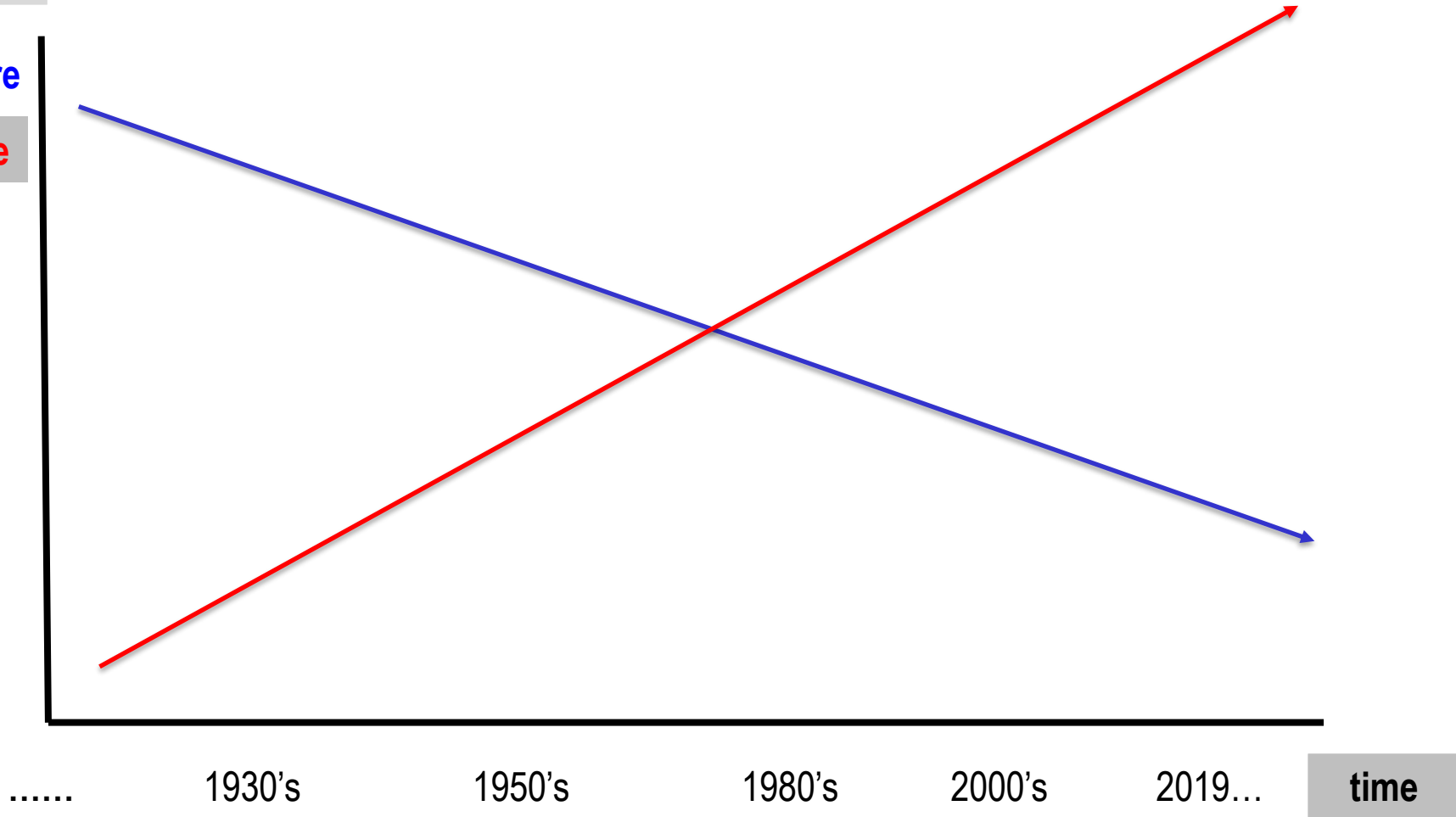


Importance of Software in the Technology (R)evolution

Cost \$
and
Size of

Hardware

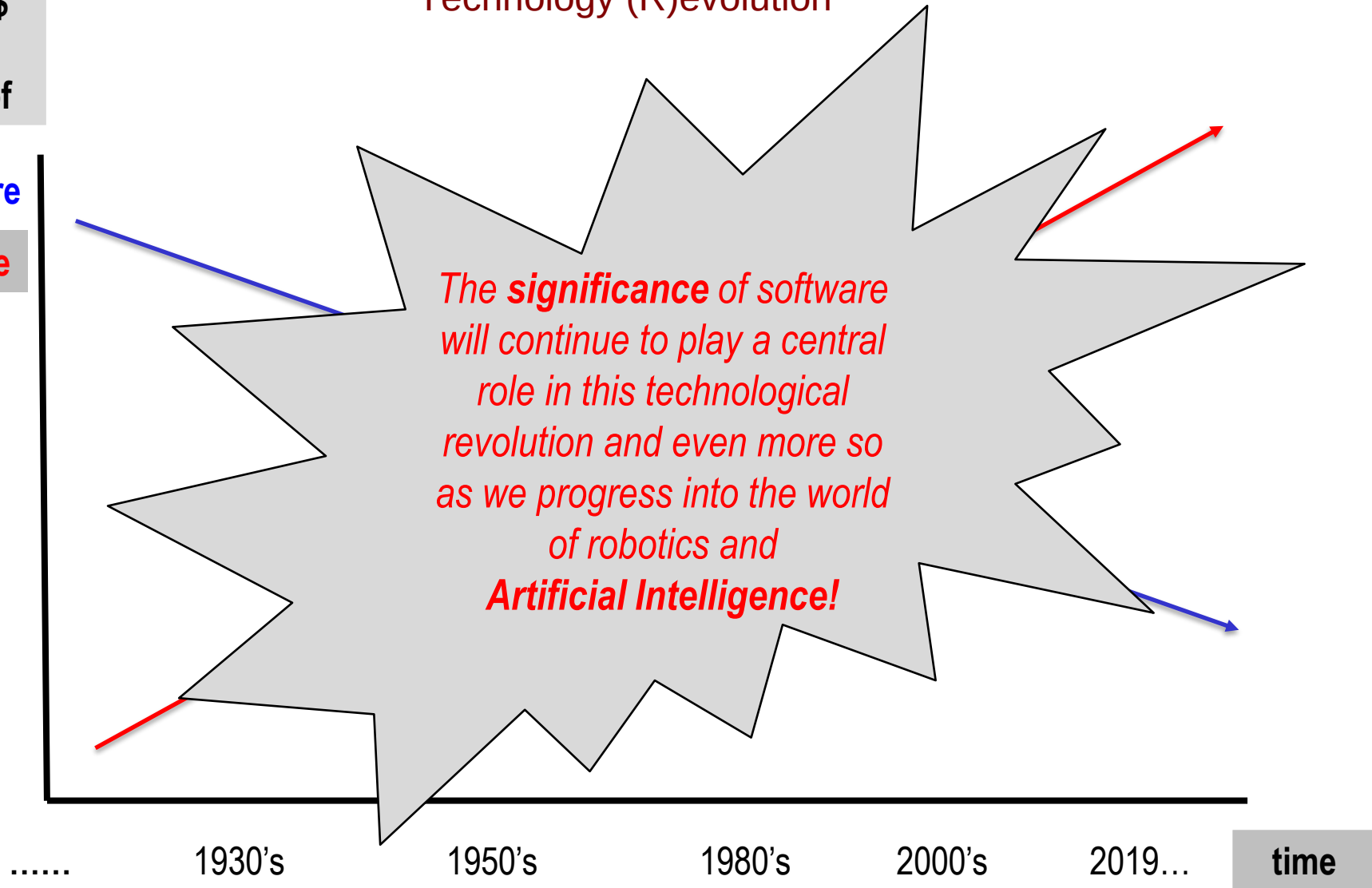
Software



Importance of Software in the Technology (R)evolution

Cost \$
and
Size of

Hardware
Software



Software and Intelligence:

Human Intelligence

- Senses
- Memory
 - Short term
 - Long term
- Reason or Logic
 - We use reason and logic to make decisions
- Repetition
 - Once we learn to do something, we can do it over and over!
- Optimization
 - Once we learn how to do something, we try to do it better!

*“You can mass-produce hardware;
you cannot mass-produce software;
you cannot mass-produce the human
mind.”*

Michio Kaku

Software and Intelligence:

Program Intelligence

- Senses ➡ Input/Output (I/O)
- Memory
 - Short term ➡ Data Types and Variables, and Data structures
 - Long term ➡ Files
- Reason or Logic ➡ Conditional Statements
 - We use reason and logic to make decisions
- Repetition ➡ Loops
 - Once we learn to do something, we can do it over and over!
- **Optimization** ➡ **Functions, OO Programming**
 - Once we learn how to do something, we try to do it better!

Motivation for Innovation

We have a problem to solve...

Understand the problem

Identify the solution

Express the solution

Motivation for Innovation

We have a problem to solve...

Understand the problem

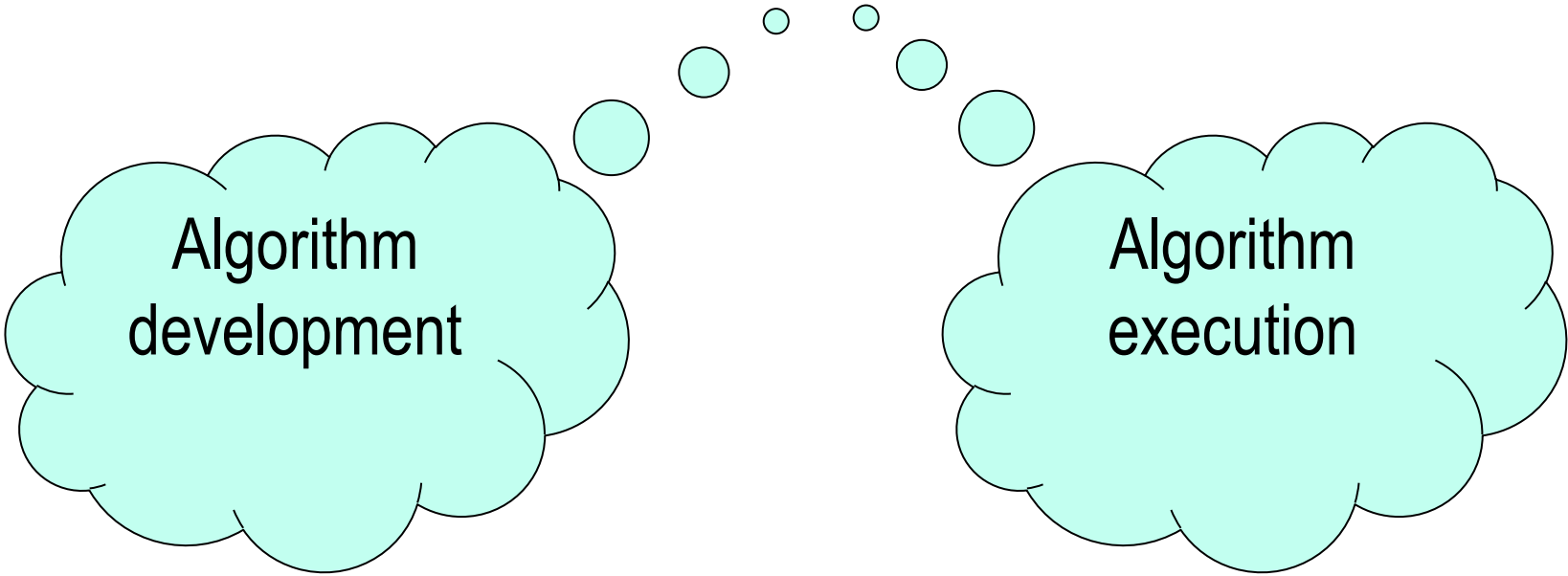
Identify the solution

Express the solution

Algorithm

Algorithm

A well ordered collection of **unambiguous** and **effectively computable** operations that when executed produces a **result** and **halts** in a finite amount of time



Algorithm
development

The diagram consists of two light blue, cloud-like shapes. The left cloud contains the text 'Algorithm development'. The right cloud contains the text 'Algorithm execution'. A series of seven small, light blue circles connects the two clouds, starting from the top of the left cloud and ending at the top of the right cloud, suggesting a flow or relationship between the two processes.

Algorithm
execution

What do each of these *advancements* or *milestones* have in common?

Jacquard's loom
Analytical Engine
Hollerith's Machines
The Turing Machine

·
·

The Algorithm

*They solved the problem by creating
an **abstract** model of the problem!*

Abstraction

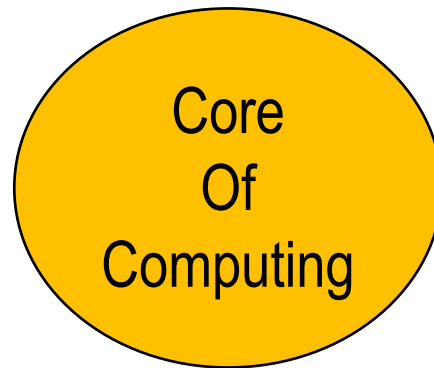
“The essence of abstraction is preserving information that is relevant in a given context and **forgetting** information that is irrelevant in that context”

John V Guttag

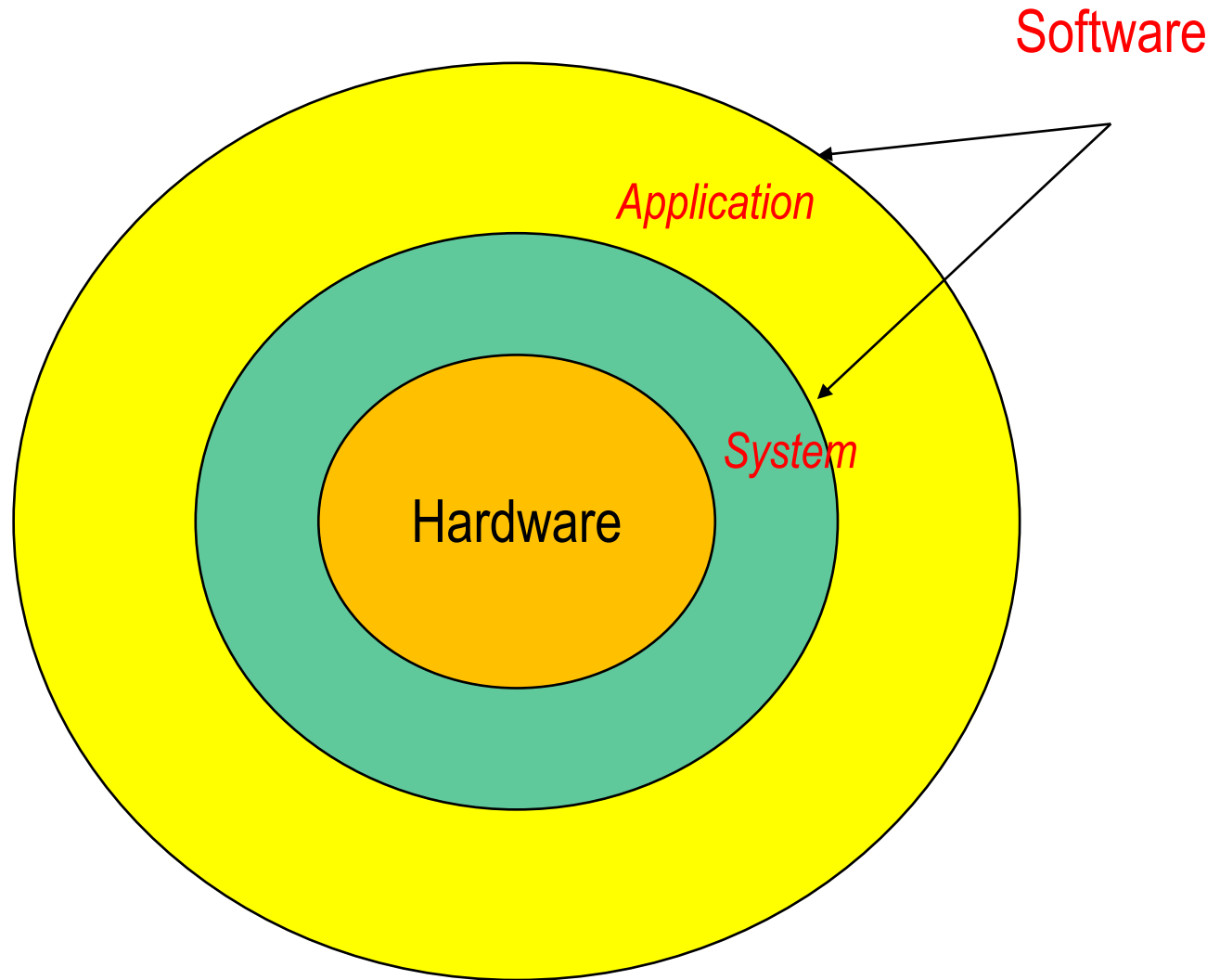
“Being abstract is something profoundly different from being vague... The purpose of abstraction is not to be vague, but to create a new semantic level in which one can be absolutely precise. --

Edsger Dijkstra

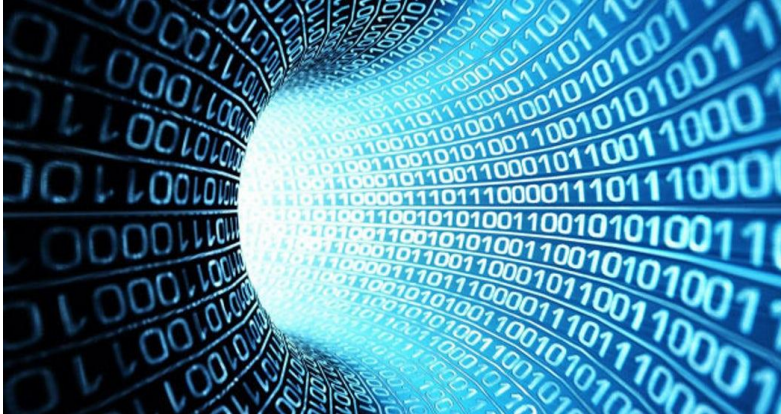
Power of Abstraction



Power of Abstraction



Abstraction in Software...



Machine Language

The only language a computer understands!

- Lowest level representation of an instruction set that a given computer can execute.
- Instructions must be executed directly by a computer's CPU
- Machine or Architecture dependent. You need to know the architecture, program registers, individual bit patterns, etc

Abstraction in Software...

```
Line No.
1      .data
2      .align 4
3      A: .long 6, -2, 3, 7, -1
4      .comm min,4,4
5      .comm max,4,4
6      .section .rodata
7      fmt: .string "min = %d, max = %d\n"
8      .text
9      .globl main
10     main:                                     # instruction size
11         pushl %ebp                           # 1 byte
12         movl %esp, %ebp                       # 2 bytes
13         movl A, %eax                          # 5 bytes
14         movl %eax, min                       # 5 bytes
15         movl %eax, max                       # 5 bytes
16         movl $1, %ecx                        # 5 bytes
17     loop:
18         cmpl $5, %ecx                         # 3 bytes
19         jge done                             # 2 bytes
20         movl A(,%ecx,4),%eax                  # 7 bytes
21         cmpl min, %eax                       # 6 bytes
22         jge next1                            # 2 bytes
23         movl %eax, min                       # 5 bytes
24     next1:
25         cmpl max, %eax                       # 6 bytes
26         jle next2                            # 2 bytes
27         movl %eax, max                       # 5 bytes
28     next2:
29         incl %ecx                           # 1 byte
30         jmp loop                            # 2 bytes
31     done:
32         pushl max                           # 6 bytes
33         pushl min                           # 6 bytes
34         pushl $fmt                          # 5 bytes
35         call printf                         # 5 bytes
36         addl $12, %esp                      # 3 bytes
37         movl $0, %eax                       # 5 bytes
38         leave                                # 1 byte
39         ret                                # 1 byte
```

Assembly
Language

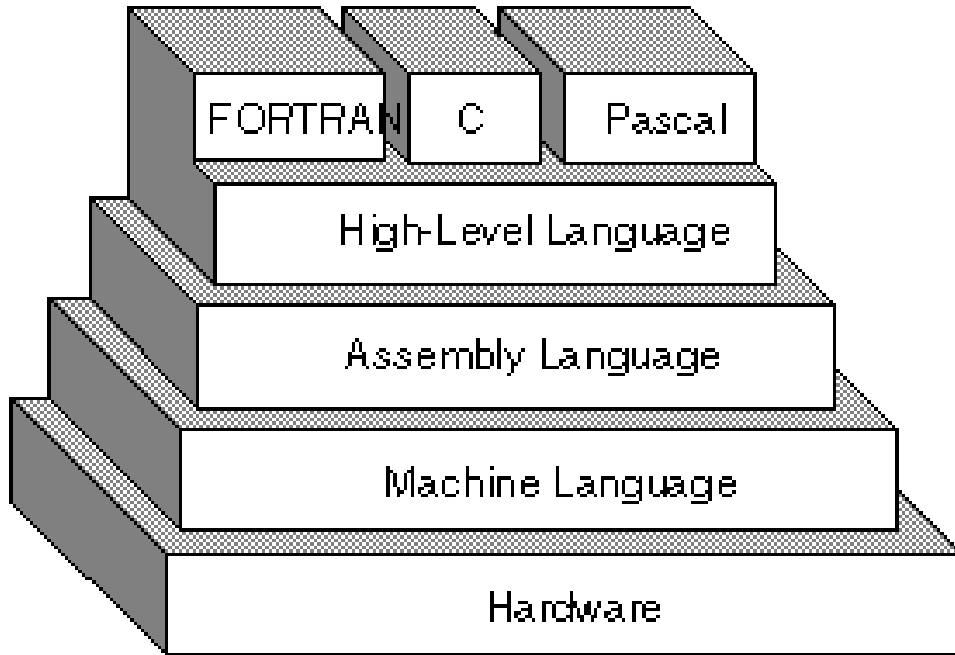
A higher level *machine* language!

Assembly
Language Code

Assembler

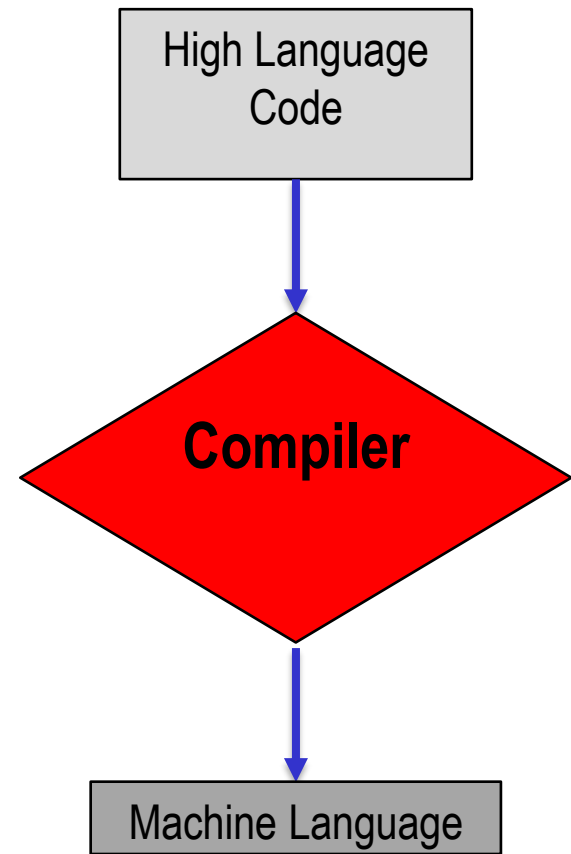
Machine Language

Abstraction in Software...



High Level
Language

A higher level *assembly* language!



Dijkstra's model

A program is broken down into logical “sub” modules that each have **one point of entry** and **one point of exit...**

- We can treat each “sub” module as an independent entity that can be called or invoked by another programs.
- Facilitates a **modular** approach to designing programs.
- “Sub” program become building blocks to building large scale systems.

Limitations to Structured Programming

The *data* and the *functions* that process the data are **independent** of each other.

Object Oriented Programming:

an evolution of Structured Programming

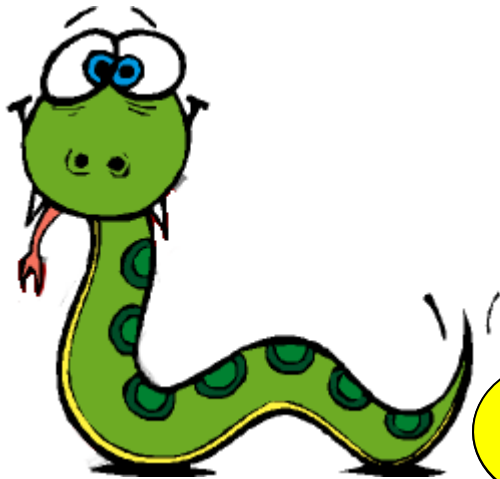
*Couples the **logic** being performed with the data it is being performed on to create a physical entity that models a real life object.*

Object Oriented programming allows this coupling to be defined implicitly within the language!

In the world of OO the buildings blocks are no longer the code modules, but the physical entities or objects that are created!

Interpreted
language
implementation.

C++



Compiled
language
implementation...